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**Computational Chemistry: Engineering Safe Drinking Water with EPANET and WNTR**

The Hook:

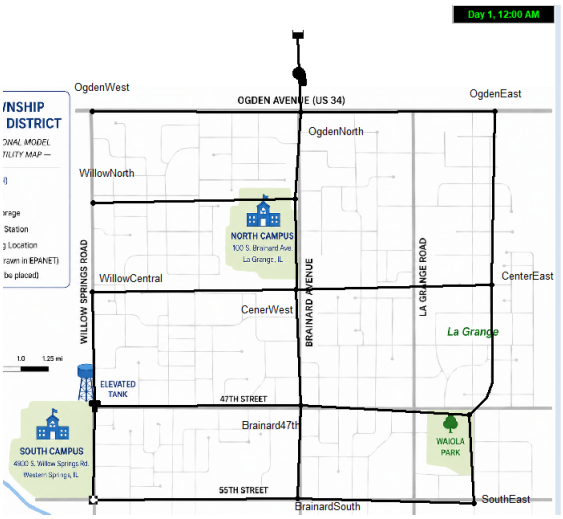
Students pass around water, pool water and complaint water and are told the following:

- Smell each sample in the scientifically accepted way. (review wafting if needed)
- North campus receives 10x the complaints about water odor than North.

Project the Water network on the board.

Discuss

- Water flow
- Dead ends
- Where water leaves the system
- Where water empties the system
- Full pipes (pressure/head)
- Water use as a function of time
- Function of the tank.
- Nodes as big systems (Strip Mall) rather than one house or one faucet.




In a large open space (outdoors, gym floor, lobby area) have the map taped on the floor with nodes labelled (See images pages 6-18) and place beakers with a water demand. Willow North, Willow Central 20 mL. North Campus, South Campus, Ogden North (Park Jr. High) are the same 50 mL. Commercial areas are 30 mL (Ogden East, Center East, South East, Brainard South), Waiola Park is the smallest 10 mL. Tank receives 100 mL for first 30 s, but gives 100 mL for second 30 s. Students start at the reservoir with a 400 mL beaker full of water. Have them move and pour some water as they pass a node. Don't tell them where to go. Let them move for 60 s.

Discuss:

- How much water was given and where.
- Amount given different times of day. (Tank here, but other spots as well)
- Dead ends
- Loops
- Where you got stuck with water
    - What happens to water stuck there?
- Where you lost your water.
- When can water flow in opposite directions?

Return to classroom and project Computational Model
https://drive.google.com/file/d/1WwLg_Do3DBeVpy4bHb4CMgwqBhlsGpy3/view?usp=drive_link

The Question:

How can an area provide enough chlorine to keep water safe while minimizing smell? (safe is 0.40 mg/L and smelly is over 1.80 mg/L)

Discuss: Students generate methods to fix the water issue. Generate these topics:

- Storage Tank size – Diameter (20, 25, 30-base, 40, 50)
- Pipe material -- PVC, Lined Iron, Unlined Iron, Old Unlined Iron, Wood
- Network design -- More loops or fewer loops

GROUP WORK – Tables of four. Different variables assigned. No graphs yet. Students hypothesize which will work best or if there will be an impact at all. Once consensus is reached, they get the data. **class of 24 will have two groups for each variable.**

Computationally modelled EPANET data sets:

- Data Water Network Variable Tank Size
- Data Water Network Variable Pipe Reactivity
- Data Water Network Variable Network Design

Data Water Network has baseline values for chlorine and data for age and a trace of water from the tank and the reservoir should discussion call for it.

Students now have to choose one of the designs and predict the strengths and weaknesses of the design and explain their choice in writing and orally. Written and oral responses must include either concentration, reaction rate, or both.

Presentations will be Day 2. During presentation a WNTR generated GIF of an acceptable choice runs and students then explain what is happening in the simulation. Students will not see animation until presentation begins.

Tank_40ft_Chlorine_Animation_With_Map.gif

Pipe_Material_-0.50_Animation.gif

Extra_Loops_Network_Animation.gif

Day 2

Students get five-minute combo group refresh (e.g. both pipe variable teams talk) and a two minute presentation is given. 2-3 minute presentation for each variable. Quick classroom check on what the best system is to reduce this issue.

Then disaster strikes (WNTR)

The instant our last presentation is done, our design will successfully solved the chlorine odor problem, but the teacher just receives an email. As a homecoming prank Hinsdale Central's Yacht Club will dump 50 jars of Almas Beluga Caviar worth over $500,000 into our water tower at noon today. The police are at a conference and will not be able to stop them. The caviar does not react with the pipes like chlorine. The water utility is calling us in ten minutes and expect answers to these questions:

- Which node is contaminated first?
- Which area receives the highest concentration?
- Which area stays clean the longest?
- Which customers should be warned first?
- Should customers continue using water?
- Does their "best chlorine network" also become the best emergency network?

WNTR simulation is then shared with students Caviar_Contamination_Animation.gif

(contamination occurs at noon day 2) and the following questions are passed out.

- Which pipes acted like highways?
- Which locations were protected by loops?
- Did the tank help or hurt?
- Which buildings had the most time to react?
- Did adding loops spread contamination faster?
- Is the "best chlorine network" also the most resilient network?

You are given $2 million by Hinsdale Humane Society to make a single improvement.

- add one new pipe
- enlarge the tank
- change a pipe material

Use data from both data sets to choose and explain your reasoning.

Teacher's Notes:
The contamination clearly spreads through much of the southern and central system, eventually reaching North Campus by Hour 54, but several locations never appear in your sampled outputs at all, especially the Ogden-area nodes. That means the contamination is following specific hydraulic pathways rather than simply diffusing everywhere.
A few strong takeaways students could generate:

- The tank is not just a storage location; it becomes a repeated contamination source each time contaminated water leaves it.
- North Campus is reached later, so it has more response time than South Campus and the central junctions.
- South Campus, Willow Central, and Brainard/47th are hit early and repeatedly, so they should be warned first.
- Loops help move water, but they can also spread contamination farther.
- Some neighborhoods are hydraulically protected under this demand pattern because contaminated water never travels through their part of the network during the 120-hour period.
- The contamination does not simply fade away smoothly. It reappears in waves when the tank starts draining again.
- The most surprising pattern is that North Campus becomes contaminated at Hour 54, drops out of the printed list at Hour 60, and then rises again by Hour 72. That suggests changing flow direction and repeated tank cycling—not a single plume moving steadily north.

NOTE: Chlorine source is treatment at the reservoir with initial quality of 1. Tank is a booster with initial quality of 1.
Prior knowledge:

- Solution parts
- Concentration mg/L compared to Molarity
- Dilutions
- Reaction rates (age of water)
- Chemical reactions

**NGSS Performance Expectations:**

HS-ETS1-2
Design a solution to a complex real-world problem by breaking it down into smaller, more manageable problems that can be solved through engineering.

HS-ETS1-3

Evaluate a solution to a complex real-world problem based on prioritized criteria and trade-offs that account for a range of constraints, including cost, safety, reliability, and aesthetics as well as possible social, cultural, and environmental impacts.

HS-ETS1-4
Use a computer simulation to model the impact of proposed solutions to a complex real-world problem with numerous criteria and constraints on interactions within and between systems relevant to the problem.

**NGSS Science and Engineering Practices:**

Developing and Using Models

Students analyze computational models of drinking water distribution systems and interpret simulation outputs.

Analyzing and Interpreting Data

Students evaluate chlorine concentration data, contaminant transport, and engineering performance across multiple system designs.

Constructing Explanations and Designing Solutions

Students explain observed water quality patterns and recommend engineering improvements supported by simulation evidence.

Engaging in Argument from Evidence

Students defend their selected network design and evaluate whether it remains the best design during an emergency contamination event.

**Cross Cutting Concepts:**

Systems and System Models

The drinking water distribution network is analyzed as an interconnected engineered system.

Cause and Effect

Students investigate how changes in infrastructure affect chlorine residuals, contaminant transport, and water quality.

Stability and Change

Students examine how normal operation differs from emergency conditions and evaluate system resilience over time.

**Disciplinary Core Ideas:**

ETS1.A – Defining and Delimiting Engineering Problems

Criteria and constraints guide engineering design decisions.

ETS1.B – Developing Possible Solutions

Multiple design alternatives are evaluated using computational models.

ETS1.C – Optimizing the Design Solution

Students compare competing designs and recommend improvements based on evidence.

PS1.B – Chemical Reactions

Chlorine reacts with pipe materials and decays over time, affecting residual concentration throughout the distribution system.